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Objective

As a final-year Artificial Intelligence engineering student at ENSIA, I am seeking a graduation project supervisor and topic proposal within the fields of AI, machine learning, and intelligent systems. My goal is to contribute to an academically rigorous project that blends research with practical implementation, leveraging my skills in NLP, reinforcement learning, computer vision, and enterprise software development.

Education

ENSIA – National School of Artificial Intelligence (Final Year)

Specialization: Artificial Intelligence & Intelligent Systems

Technical Skills

- **Programming & Development:** Python, Java (Spring Boot), Dart/Flutter, PHP, HTML, CSS, JavaScript, Flask.
- **AI & Machine Learning:** Supervised/unsupervised learning, deep learning (ANNs, CNNs), reinforcement learning (A3C), predictive modeling, time series analysis.
- **NLP & Text Processing:** Corpus building, text style transfer, semantic search, feature engineering.
- **Computer Vision & Speech Processing:** Image preprocessing, feature extraction, classification; signal analysis.
- **Networking & Security:** WLAN security monitoring, intrusion detection, packet analysis, Grafana dashboards.
- **Data Science Tools:** Pandas, NumPy, SciPy, Scikit-learn, Matplotlib, Seaborn, R.
- **Databases & Big Data:** MySQL, Firebase, InfluxDB, ChromaDB, MongoDB, Cassandra; Hadoop ecosystem (HDFS, MapReduce, YARN), Apache Spark, Hive; batch & stream processing (Kafka, Flink); scalable ML with Spark MLlib; cloud-based big data services (AWS, Google Cloud, Azure).

Analytical and Theoretical Skills

- Solid foundation in AI principles including neural networks, optimization techniques, and stochastic modeling.

- Experience in data preprocessing, feature engineering, and algorithmic problem-solving.
- Capable of bridging theory with implementation, from mathematical modeling to production-ready systems.
- Ability to integrate concepts from multiple AI subfields (NLP, RL, CV, HPC) into cohesive project solutions.

Projects

Jisser al-Logha – Arabic Literary Style Transfer (NLP)

Technologies: Python, Hugging Face Transformers, LoRA fine-tuning, SentenceTransformers, ChromaDB.

Description: Developed a hybrid rule-based and ML system to convert text between Classical and Modern Arabic, supported by a synthetic parallel corpus (~7k pairs) and semantic search functionality.

AI-Powered WLAN Security Monitoring System (Wireless Networks + ML)

Technologies: Python, Scapy, Random Forest, InfluxDB, Grafana, Docker.

Description: Designed and deployed a real-time intrusion detection system for WLAN attacks (deauthentication, disassociation, rogue AP) with 99% classification accuracy and live dashboards.

A3C Reinforcement Learning for Traffic Signal Control (Reinforcement Learning)

Technologies: Python, SUMO, TensorBoard.

Description: Implemented a CNN-based A3C agent to control traffic lights under partial detection (40% vehicle visibility), achieving up to 72% delay reduction compared to fixed-time control.

Predicting Aqua Solubility (Data Mining)

Technologies: Python, feature engineering, predictive modeling.

Description: Built regression models for compound water solubility prediction with emphasis on data preprocessing.

Time Series Classification of ECG Signals (Time Series + CV)

Technologies: Python, ANNs, feature extraction.

Description: Classified ECG signals using image transformation and engineered time series features.

Loan Scoring (Loan Eligibility) (ML + Web)

Technologies: Python, ML, Web Development.

Description: Predicted loan eligibility using real Algerian bank data.

BlediTourism (Web Development)

Technologies: PHP, HTML, CSS, JavaScript.

Description: Tourism management platform integrating user-generated content and service provider tools.

Drivera (Web Development)

Technologies: Web Development, Database Schema Design, Notification Systems.

Description: Driving school management system with student, tutor, and vehicle tracking.

Nim Game (AI + Game Dev)

Technologies: Python, Tkinter, Minimax Algorithm, GUI Design.

Description: Implemented the Minimax algorithm in a Python/Tkinter-based strategy game.

Unimate (Mobile Development)

Technologies: Flutter, Dart, Flask, Firebase.

Description: Educational social platform for Algerian students with forums, messaging, and resource sharing.